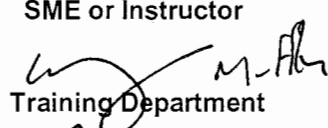
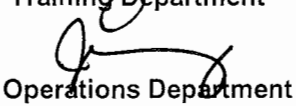


**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM		
SYSTEM:	Rod Control		
TASK:	Perform a Control Rod Operability Verification		
TASK NUMBER:	N0010070101		
JPM NUMBER:	15-01 NRC Sim a		
ALTERNATE PATH:	☒	K/A NUMBER:	001 A4.03
APPLICABILITY:		IMPORTANCE FACTOR:	
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator		
REFERENCES:	S2.OP-ST.RCS-0001, Rev. 22 (checked 8-3-16)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	<u>10 min</u>		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>		
Developed By:	G Gauding Instructor	Date:	8-3-16
Validated By:	M Lutek / W Russell SME or Instructor	Date:	8-22-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE
JOB PERFORMANCE MEASURE
SIMULATOR SETUP INSTRUCTIONS**

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

TASK NUMBER: N0010070101

SIMULATOR IC: **IC-251** in 15-01 NRC JPMs

MALFUNCTIONS
REQUIRED: RT-1 will trip 21 SGFP
MALF: BF0105A, 21 STM GEN FEED PUMP TRIP, Final Value: 3

OVERRIDES
REQUIRED: None

SPECIAL
INSTRUCTIONS: None

Note to Evaluator: OP-AP-300-1001 required a peer check for all non-emergency manual rod motion. Do NOT correct operator if a mistake is made when performing a peer check. For all required peer checks simply state "Peer check SAT".

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

TASK NUMBER: N0010070101

INITIAL CONDITIONS:

Unit 2 is operating at steady state, 100% power. Power has been at 100% for 30 days.

INITIATING CUE:

- You are directed to perform S2.OP-ST.RCS-0001, Reactivity Control System Rod Control Assemblies.
- A Maintenance Technician is stationed at the Rod Control System Power Cabinets.
- Communications are established via plant page between the Control Room and Jeff Stevens, NEO, at the Rod Control System Power Cabinets.
- The CRS has directed that steps 5.1.2 through 5.1.9 are to be performed in order.
- The CRS directs that 15 steps of rod insertion are to be performed for each bank to ensure that each rod moves at least 10 steps.
- Initial Control Rod Positions are:

SDA		SDB		SDC		SDD	
F-04	229	G-03	229	E-03	229	C-05	229
D-10	228	C-09	227	C-11	223	E-13	223
K-12	227	J-13	227	L-13	224	N-11	224
M-06	227	N-07	229	N-05	224	L-03	227
B-04	230	J-02	223				
D-14	224	C-06	223				
P-12	229	G-13	227				
M-02	230	N-09	230				

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

Operator trips the Rx when Shutdown Bank B is inserted in the core and a SGFP trips.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Provide marked up copy of S2.OP-ST.RCS-0001, Rev. 22, with Prerequisite Sections 2.1-2.8 filled out.	Reviews Prerequisites.		
		Provide copy of OP-AP-300-1001, PWR Control Rod Movement Requirements.	Reviews OP-AP-300-1001, PWR Control Rod Movement Requirements.		
			Reviews and signs Precautions and Limitations. Cue: <u>IF</u> ASKED, state that no special contingency actions have been briefed other than what is in the P&L's.		
	5.1.1	IF this surveillance is to be performed....	Determines step is N/A based on steady state 100% operation.		
	5.1.2	PERFORM Attachment 2, Rod Control System Voltage Checks as follows: ___ A. PLACE Rod Bank Selector Switch to "MANUAL". ___ B. Direct Maintenance to COMPLETE Attachment 2, Rod Control System Voltage Checks.	Performs Attachment 2 Rod Control System Voltage Checks by: A. Places Rod Bank Selector Switch to manual. B. Directs Direct Maintenance to complete Attachment 2, Rod Control System Voltage Checks. Cue: Maintenance reports Attachment 2 Rod Control System Voltage Checks are complete SAT.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	Note	— The Shutdown Rod Insertion Limit (TS 3.1.3.4) is not applicable during the performance of this test for rods actually undergoing testing. — Steps 5.1.3 through 5.1.10 may be performed in any order.	Initials both notes.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.1.3	PERFORM Shutdown Bank A testing as follows: A. <u>IF</u> the Rx is critical, <u>THEN</u> ENSURE T_{avg} is within $\pm 1^{\circ}\text{F}$ of T_{ref}. B. PLACE Bank Selector Switch in the "SBA" position. C. ENSURE GRP. SELECT "C" lights are illuminated on Rod Control System Power Cabinets 21AC and 22AC. D. Maneuver Shutdown Bank A at least 10 steps in any one direction. E. ENSURE each rod in Shutdown Bank A indicated rod movement of at least 10 steps. F. RECORD Shutdown Bank A "Test Results" by initialing in Attachment 1. G. RESTORE Shutdown Bank A to pre-test condition.	Checks Tave/Tref recorder on 2RP4 and notes T_{avg} is within 1°F of T_{ref}. Rotates Bank Selector Switch fully counter clockwise to the SBA position. Contacts NEO and directs him to verify GRP. SELECT "C" lights are illuminated on Rod Control System Power Cabinets 21AC and 22AC. <u>Cue:</u> GRP SELECT "C" lights are illuminated at Rod Control System Power Cabinets 21AC and 22AC. IAW OP-AA-300-1001, prior to rod movement, operator states initial Shutdown Bank A position, target control rod position, and direction of rod movement, then inserts SBA at least 10 steps to stated target position. Verifies rod movement by viewing rod position on P-250 computer. Initials SAT for Groups 1 and 2 for SBA on Att 1. Operator states initial Shutdown Bank A position, target control rod position, and direction of rod movement, then withdraws SBA to stated target ARO position. Continuous rod withdrawal is acceptable.		

OPERATOR TRAINING PROGRAM

JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.1.4	PERFORM Shutdown Bank B testing as follows: A. <u>IF</u> the Rx is critical, <u>THEN ENSURE</u> T_{avg} is within $\pm 1^{\circ}\text{F}$ of T_{ref}. B. PLACE Bank Selector Switch in the "SBB" position. C. ENSURE GRP. SELECT "C" lights are illuminated on Rod Control System Power Cabinets 21BD and 22BD. D. MANUEVER Shutdown Bank B at least 10 steps in any one direction. E. ENSURE each rod in Shutdown Bank B indicated rod movement of at least 10 steps.	Checks Tave/Tref recorder on 2RP4 and notes Tavg is within 1°F of Tref. Rotates Bank Selector Switch clockwise to the SBB position. Contacts NEO and directs him to verify GRP. SELECT "C" lights are illuminated on Rod Control System Power Cabinets 21BD and 22BD. <u>Cue:</u> GRP SELECT "C" lights are illuminated at Rod Control System Power Cabinets 21BD and 22BD. Operator states initial Shutdown Bank B position, target control rod position, and direction of rod movement, then inserts SBB at least 10 steps to stated target position. <u>SIMULATOR OPERATOR:</u> INSERT RT-1 when the operator is checking rod position after insertion. This will trip 21 SGFP. Responds to 21 SGFP trip by tripping the Rx IAW P&L 3.13. (This is Alternate Path) Movement of the Rod Bank Selector Switch to MAN or AUTO under these conditions is incorrect.		
			When the operator has either tripped the Rx or initiated rod insertion in MAN or AUTO, state JPM is complete.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

TQ-AA-106-0303
Revision 4

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ 2/17 1. Task description and number, JPM description and number are identified.
- ✓ 2/17 2. Knowledge and Abilities (K/A) references are included.
- ✓ 2/17 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ 2/17 4. Initial setup conditions are identified.
- ✓ 2/17 5. Initiating and terminating Cues are properly identified.
- ✓ 2/17 6. Task standards identified and verified by SME review.
- ✓ 2/17 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- ✓ 2/17 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 22 Date 6/13/15
- 2/17 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- 2 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: M Lutek

Date: 8/22/16

SME/Instructor: W. Russell Russell

Date: 8/22/16

SME/Instructor: _____

Date: _____

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

INITIAL CONDITIONS:

Unit 2 is operating at steady state, 100% power. Power has been at 100% for 30 days.

INITIATING CUE:

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K-12 227	J-13 227	L-13 224	N-11 224
M-06 227	N-07 229	N-05 224	L-03 227
B-04 230	J-02 223		
D-14 224	C-06 223		
P-12 229	G-13 227		
M-02 230	N-09 230		